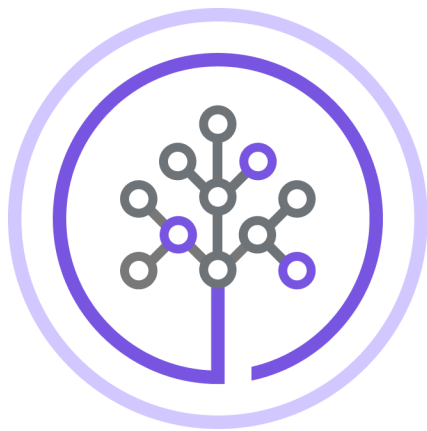


Creating Data Visualizations using ggplot

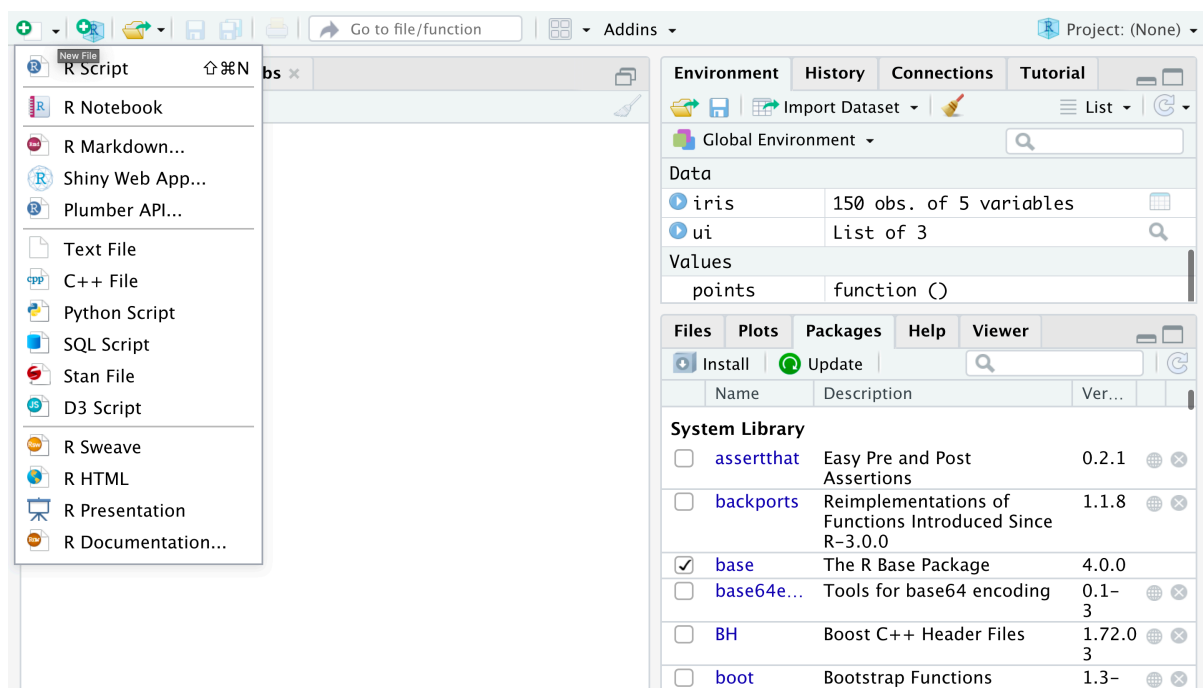


Skills Network

Objective for Exercise

We will create different data visualizations using the `ggplot` package using the inbuilt dataset in R called `mtcars`

1. Click on the `+` symbol on the top left and choose `R Script` from the menu to open a new R edit window in RStudio:



2. Read and view the first 5 rows of the Data using the following:

```
library(datasets)

#Load Data

data(mtcars)

#View first 5 rows

head(mtcars, 5)
```

3. Type this `?mtcars` to get information about the variables. This will print the information at the bottom right panel, on the `Help` tab

4. Copy and paste the following code to load the `ggplot` package and create a scatterplot of `disp` and `mpg`.

```
#load ggplot package
library(ggplot2)

#create a scatterplot of displacement (disp) and miles per gallon (mpg)

ggplot(aes(x=disp,y=mpg,),data=mtcars)+geom_point()
```

5. Use the following code to add a title.

```
#Add a title

ggplot(aes(x=disp,y=mpg,),data=mtcars)+geom_point()+ggtitle("displacement
```

6. Use the following code to change the name of the `x-axis` and `y-axis`

[plaintext](#)

```
#change axis name
```

```
ggplot(aes(x=disp,y=mpg,),data=mtcars)+geom_point()+ggtitle("displacement
```

7. Use the following to create a boxplot of the the distribution of `mpg` for the individual Engine types `vs Engine (0 = V-shaped, 1 = straight)` To do this you have to make `vs` a string or factor.

[plaintext](#)

```
#make vs a factor
```

```
mtcars$vs <- as.factor(mtcars$vs)
```

```
#create boxplot of the distribution for v-shaped and straight Engine
```

```
ggplot(aes(x=vs, y=mpg), data = mtcars) + geom_boxplot()
```

8. Add color to the boxplots to help differentiate:

[plaintext](#)

```
ggplot(aes(x=vs, y=mpg, fill = vs), data = mtcars) +  
  geom_boxplot(alpha=0.3) +  
  theme(legend.position="none")
```

9. Finally, let us create the histogram of weight `wt`.

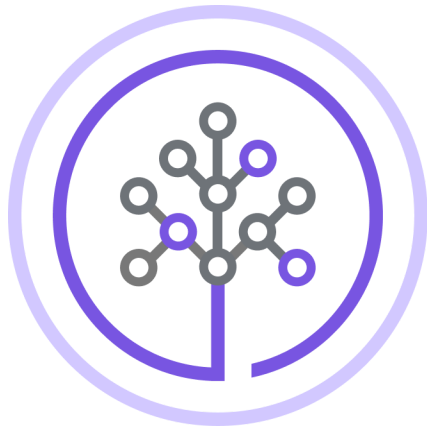
[plaintext](#)

```
ggplot(aes(x=wt),data=mtcars) + geom_histogram(binwidth=0.5)
```

This concludes this lab, we hope that you had fun!

Author(s)

Creating Data Visualizations using ggplot



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Objective for Exercise

We will create different data visualizations using the `ggplot` package using the inbuilt dataset in R called `mtcars`

1. Click on the `+` symbol on the top left and choose `R Script` from the menu to open a new R edit window in RStudio:

The screenshot shows the RStudio interface. The top-left pane displays a 'New File' menu with various file types. The top-right pane shows the 'Environment' tab with 'Global Environment' selected, displaying 'iris' (150 obs. of 5 variables) and 'ui' (List of 3). The bottom-right pane shows the 'Packages' tab with a list of installed packages including assertthat, backports, base, base64e..., BH, boot, and callr.

2. Read and view the first 5 rows of the Data using the following:

```
library(datasets)

#Load Data

data(mtcars)

#View first 5 rows

head(mtcars, 5)
```

3. Type this `?mtcars` to get information about the variables. This will print the information at the bottom right panel, on the `Help` tab
4. Copy and paste the following code to load the `ggplot` package and create a scatterplot of `disp` and `mpg`.



plaintext



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#load ggplot package
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ggplot(aes(x=disp,y=mpg,),data=mtcars)+geom_point()
```

5. Use the following code to add a title.



plaintext



```
#Add a title

ggplot(aes(x=disp,y=mpg,),data=mtcars)+geom_point()+ggtitle("displacement
```

6. Use the following code to change the name of the **x-axis** and **y-axis**



plaintext



```
#change axis name

ggplot(aes(x=disp,y=mpg,),data=mtcars)+geom_point()+ggtitle("displacement
```

7. Use the following to create a boxplot of the the distribution of **mpg** for the individual Engine types **vs Engine (0 = V-shaped, 1 = straight)** To do this you have to make **vs** a string or factor.



plaintext



```
#make vs a factor
mtcars$vs <- as.factor(mtcars$vs)

#create boxplot of the distribution for v-shaped and straight Engine

ggplot(aes(x=vs, y=mpg), data = mtcars) + geom_boxplot()
```

[plaintext](#)

```
ggplot(aes(x=vs, y=mpg, fill = vs), data = mtcars) +  
  geom_boxplot(alpha=0.3) +  
  theme(legend.position="none")
```

9. Finally, let us create the histogram of weight `wt`.

[plaintext](#)

```
ggplot(aes(x=wt), data=mtcars) + geom_histogram(binwidth=0.5)
```

This concludes this lab, we hope that you had fun!

Author(s)

[Aije Egwaikhide](#)



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